# Mathematical Induction 5.1 & 5.3

## Chapter Summary

1. Mathematical Induction
2. Examples of Proof by Mathematical Induction
3. Recursive Definitions

## Climbing an Infinite Ladder


Suppose we have an infinite ladder:

1. We can reach the first rung of the ladder. 2. If we can reach a particular rung of the ladder, then we can reach the next rung.

**The principle of Mathematic induction:** 1. Show we can reach the first step

2. Show that if we can reach step k, then we can reach step k+1

if we establish these 2 things, we know we can reach all the steps of the ladder.

This example motivates proof by mathematical induction.

# Principle of Mathematical Induction (MI)

Principle of Mathematical Induction: To prove that P(n) is true for all positive integers n, where P(n) is a propositional function, we complete these steps:

1. **Basis Step:** Show that P(1) is true.
2. **Inductive Step:** Show that $P(k) \to P(k + 1)$ is true for all positive integers k.

Mathematical induction can be expressed as the rule of inference

$$(P(1) \land \forall k\,(P(k) \to P(k + 1))) \to \forall n\, P(n),$$

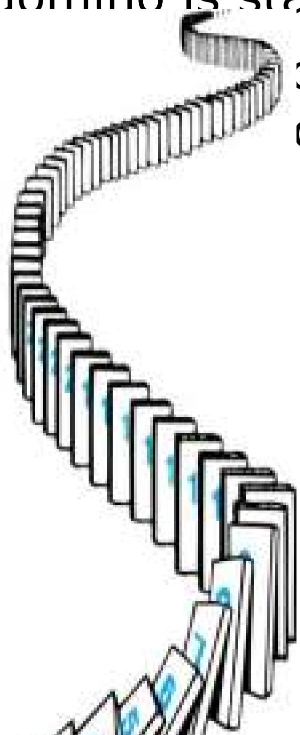
where the domain is the set of positive integers.

Proofs by mathematical induction do not always start at the integer 1. In such a case, the basis step begins at a starting point *b* where *b* is an integer. We will see examples of this soon.

# Remembering How Mathematical

# **Induction Works -** Domino Effect

Consider an infinite sequence of dominoes, labeled $1,2,3$, …, where each domino is standing.



oposition that the
cked over.

We know that the first domino is knocked down, i.e., $P(1)$ is true .

We also know that if whenever the $k$th domino is knocked over, it knocks over the $(k + 1)$st domino, i.e, $P(k) \rightarrow P(k + 1)$ is true for all positive integers $k$.

Hence, all dominos are knocked over. $P(n)$ is true for all positive integers $n$.

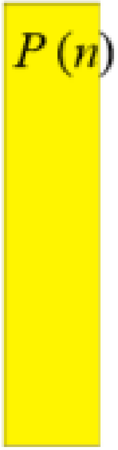
**Mathematical Induction**

Suppose we have a sequence of propositions which we would like to prove:

$P(0)$, $P(1)$, $P(2)$, $P(3)$, $P(4)$, … $P(n)$, …

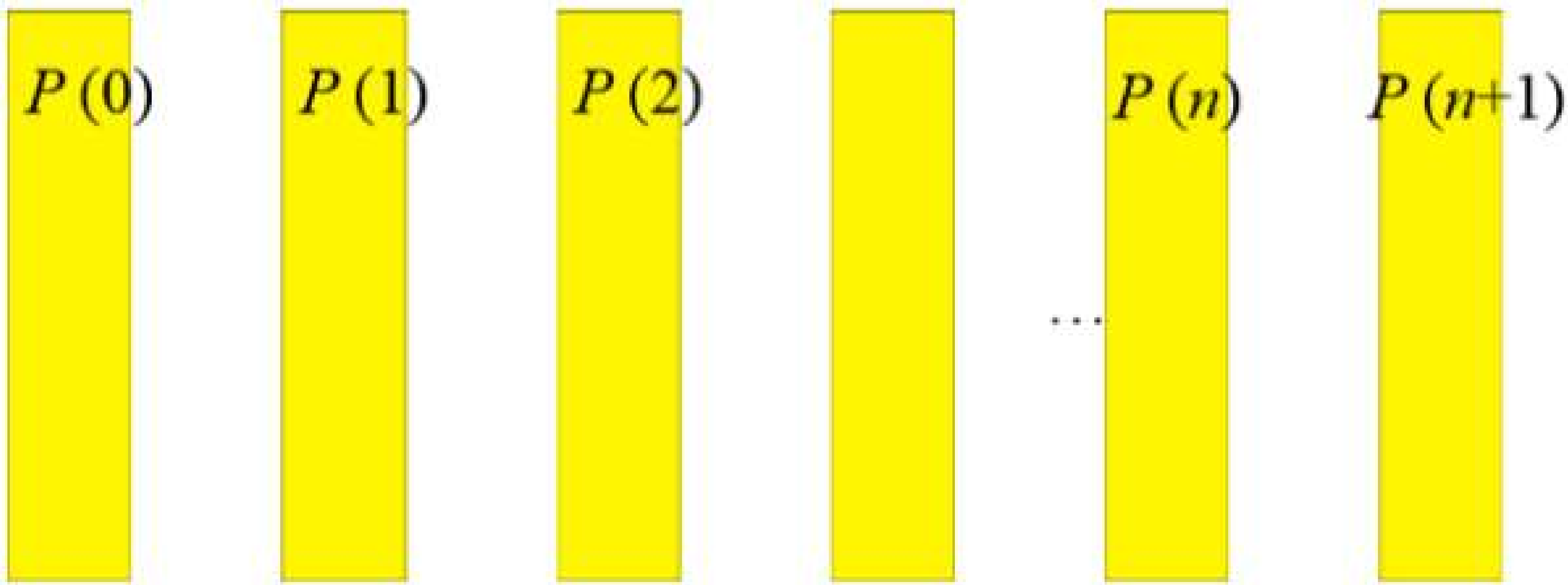
EG: $P(n)$ =

We can picture each proposition as a domino:



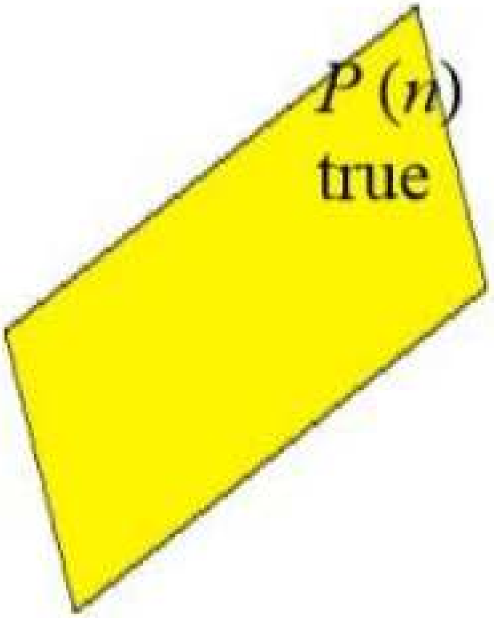

# Mathematical Induction

So sequence of propositions is a sequence of dominos.



# Mathematical Induction

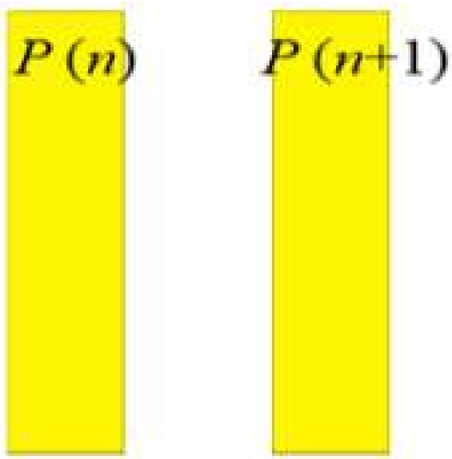
When the domino falls (to right), the corresponding proposition is considered true:



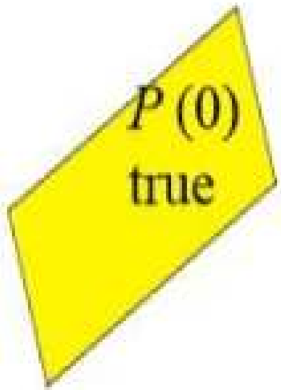

## Mathematical Induction

Suppose that the dominos
satisfy two constraints.

1) Well-positioned: If any domino falls (to right), next domino (to right) must fall also.




2) First domino has fallen to right




# Mathematical Induction

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# Conjecturing and Proving Correct a Summation Formula

**Example**: Conjecture and prove correct a formula for the sum of the first *n* positive odd integers. Then prove your conjecture.

**Solution**: We have: $1 = 1, 1 + 3 = 4, 1 + 3 + 5 = 9, 1 + 3 + 5 + 7 = 16, 1 + 3 + 5 + 7 + 9 = 25.$ • *We can conjecture that the sum of the first n positive odd integers is* $n^2$, $1 + 3 + 5 + \cdots + (2n - 1) = n^2$.

- *We prove the conjecture is proved correct with mathematical induction.*

1. ***BASIS STEP****: P*(1) *is true since* $1^2 = 1$.
2. INDUCTIVE STEP: *$P(k) \rightarrow P(k + 1)$ for every positive integer k.* Assume the inductive hypothesis holds and then show that *P*(*k*) holds has well. Inductive

Hypothesis: $1 + 3 + 5 + \cdots + (2k - 1) = k^2$

- *So, assuming P(k), it follows that:*

$$1 + 3 + 5 + \cdots + (2k - 1) + (2k + 1) = [1 + 3 + 5 + \cdots + (2k - 1)] + (2k + 1)$$
$$= k^2 + (2k + 1) \text{ (by the inductive hypothesis)}$$
$$= k^2 + 2k + 1$$
$$= (k + 1)^2$$

- *Hence, we have shown that P(k + 1) follows from P(k). Therefore the sum of the first n positive odd integers is $n^2$.*

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**Proof:**

## MathematicalInduction Example-p.318

# MathematicalInduction Example-p.318

## 5.3. Recursive Definitions

# Recursive Definitions& Induction

# Recursively Defined Functions

**Definition**: A *recursive* or *inductive definition* of a function consists of two steps.

1. ***BASIS STEP****: Specify the value of the function at zero.*
2. ***RECURSIVE STEP****: Give a rule for finding its value at an integer from its values at*

# Recursively Defined Functions

**Example1:** Suppose $f$ is defined by:

## Example 2

## Example 3

## Example 3

## Example 4

# Example 4

# Example 5

# Example 5